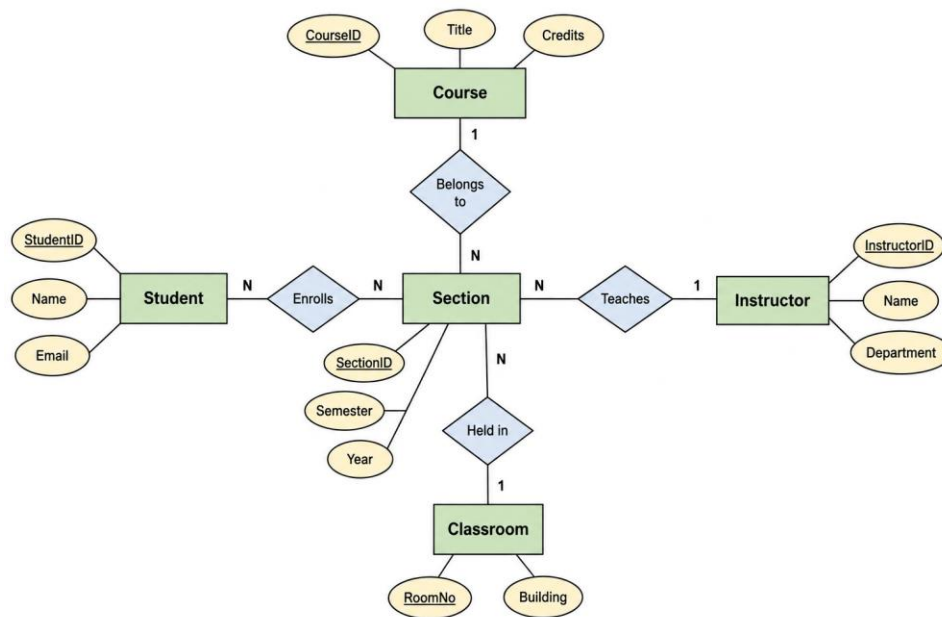


DBMS LAB ASSIGNMENT - 1

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24MAB0A32

ER Diagram – 1



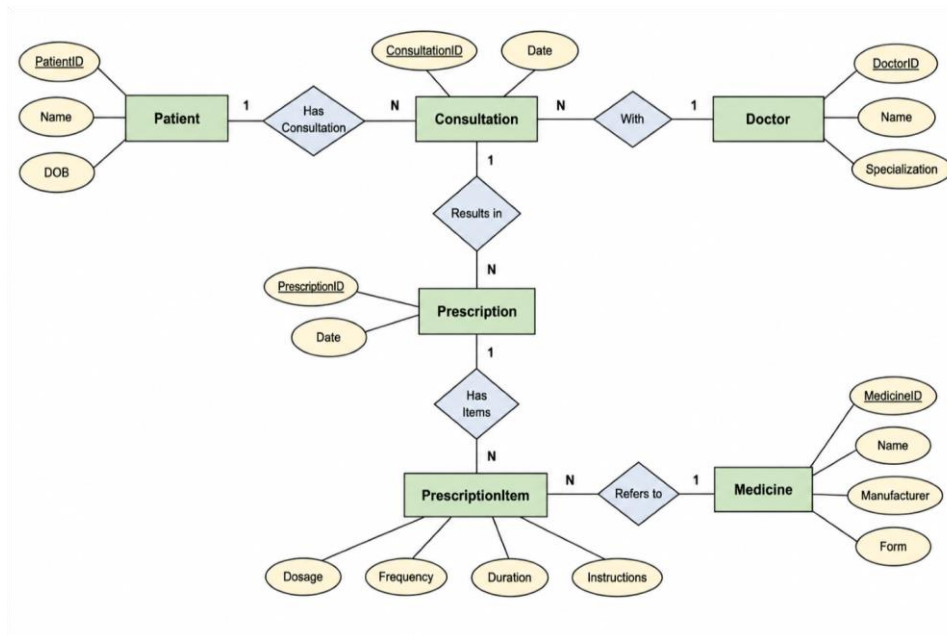
Flaws:

According to the given rules, a course can have multiple sections, and each section can have a different instructor. So, we need to add an entity called “**Section**”. A student enrolls in a section, and a course can have multiple sections.

1. **Semester and Year** should be attributes of **Section** because the same course can be offered in different sections, semesters, and years. If we keep them as attributes of Course, we may lose information about when and in which semester the course was offered.
2. A **Student** should enroll in a **Section** instead of directly enrolling in a Course because students register for a particular section of a course.
3. Different sections of the same **Course** can be taught by different **Instructors**. Therefore, the **Instructor** should teach a Section rather than the Course directly.

4. The **Classroom** can be different for different sections. Therefore, a particular **Section** should be assigned to a Classroom instead of assigning the Classroom directly to the Course.

ER Diagram – 2



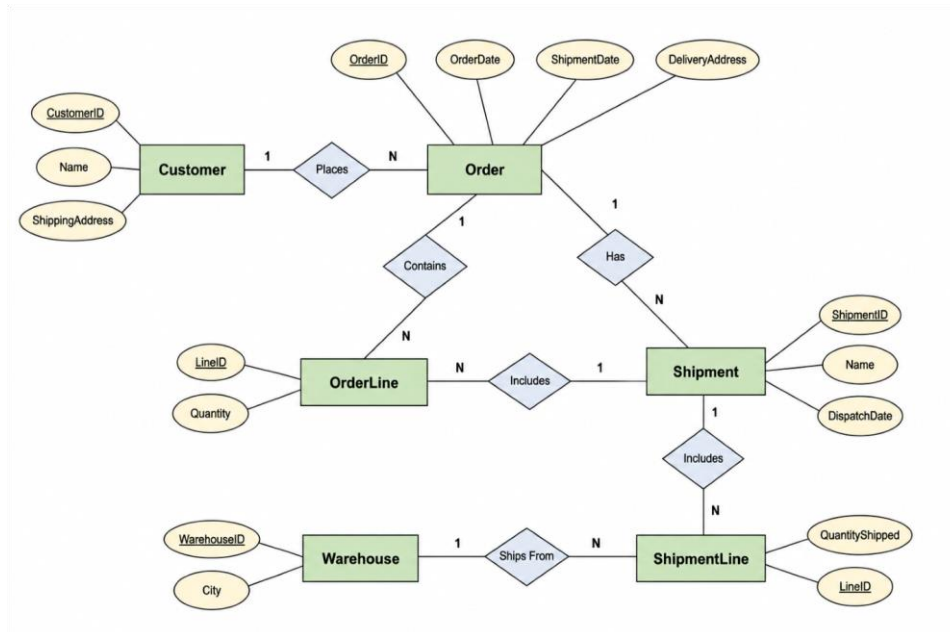
Flaws:

We need to maintain the consultation history because a patient may visit the same doctor multiple times. So, we should add an entity called “**Consultation**”. Each consultation can result in a “**Prescription**”, and the prescription can contain “**PrescriptionItem**” because the dosage belongs to the PrescriptionItem and not directly to the Medicine.

1. **Consultation** should be an entity because a patient may consult the same doctor many times. This helps us maintain the complete consultation history.
2. **Prescription** should also be an entity because it helps us keep the prescription history whenever a patient consults a doctor.
3. **PrescriptionItem** should be an entity to store the dosage, because the same medicine may be given in different dosages to different patients.

4. For instructions that depend on time, **Frequency** and **Duration** can be added as attributes of **PrescriptionItem**.

ER Diagram – 3



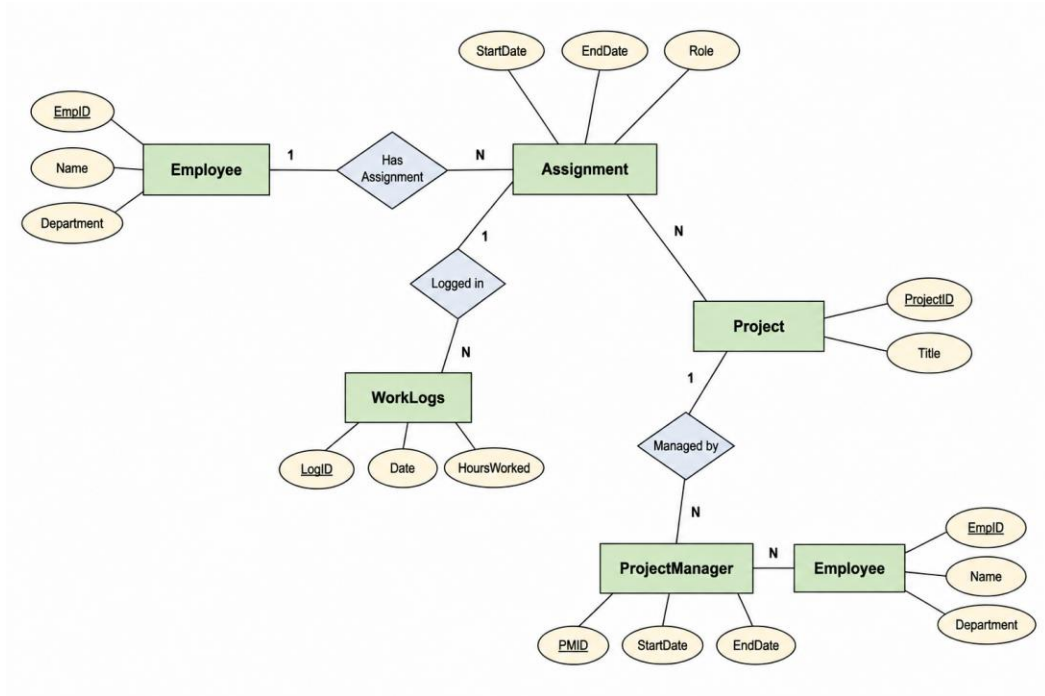
Flaws:

The “**OrderLine**” entity is missing and should be added because the quantity is related to a particular OrderLine and not directly to the Product. Also, we need another entity called “**ShipmentLine**” to handle partially shipped orders.

1. **Address** should be an attribute of **Order** instead of Customer because a customer may use different addresses for different orders.
2. **OrderLine** should be an entity because the quantity belongs to a particular OrderLine and not to the Product.
3. **ShipmentLine** should be added to keep track of OrderLines that are shipped partially.

4. **Shipment** should have its own **Dispatch Date** so that multiple shipments can be made for the same order.

ER Diagram – 4



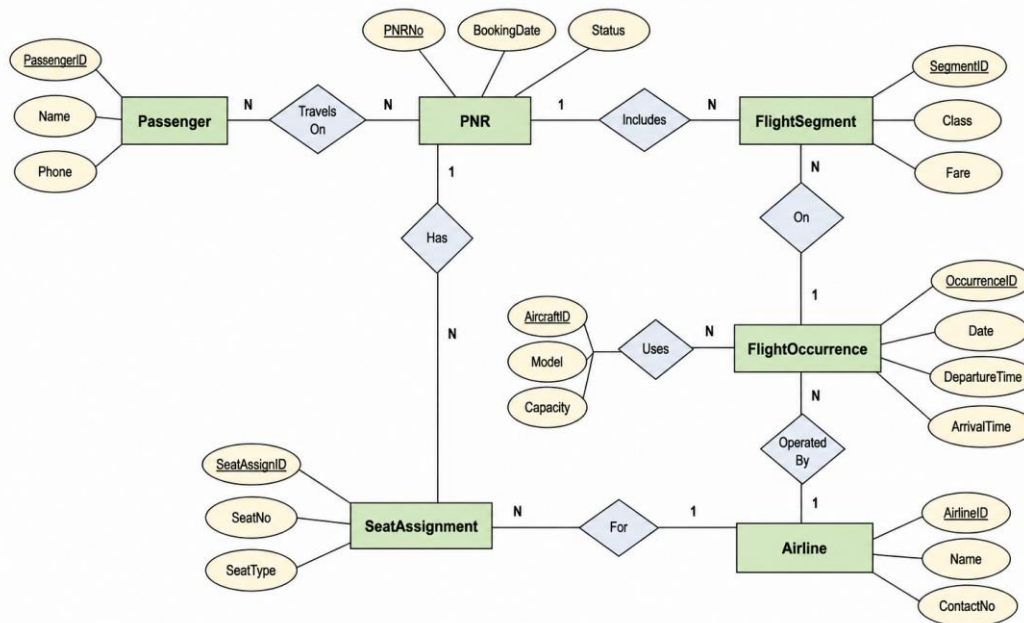
Flaws:

The relationship between **Employee** and **Project** should be converted into an entity so that we can properly maintain the history of employees working on different projects.

1. There should be an **Assignment** entity to store the role of a particular employee in a project.

2. The **Works_On** relationship cannot properly show when an employee joins, leaves, or rejoins a project. Therefore, **Start Date** and **End Date** should be added as attributes of Assignment.
3. Since the working hours are recorded for each employee-project assignment, they cannot be stored directly in the **Project** entity. Also, hours need to be recorded for different time periods, so we can create a **WorkLogs** entity to store the hours worked for each assignment during a particular time period.
4. **Start Date** and **End Date** should also be added to the **MANAGED_BY** relationship because the manager of a project may change over time.

ER Diagram – 5



Flaws:

The current ER diagram does not properly show how **bookings, flight occurrences, aircraft changes, and seat assignments** are handled. So, some additional entities are required to represent these details correctly.

1. There should be a **PNR** entity because one PNR can have multiple passengers and can include multiple flight segments.
2. There should be a **FlightOccurrence** entity because the same flight number can operate on different dates.
3. The **Aircraft** entity should be connected to **FlightOccurrence** because the aircraft assigned to a particular flight may be different for different occurrences.
4. A **SeatAssignment** entity should be added to store different seat numbers for the same passenger when they travel on different flight segments.

